OBJECT ORIENTED PROGRAMMING WITH C++

ASSIGNMENT 2 – CO2

CO2 Coverage: Design C++ programs by applying functions and object-oriented concepts – function prototyping, call by reference, return by reference, inline functions, default arguments, function overloading, specifying a class, member functions, access specifiers, arrays within a class, static data members and member functions, arrays of objects and friend functions.

Nature of Assignment : Individual
Duration : 1 Week
Total Marks : 50
Submission Date : _____

Bloom's Levels Assessed: K5 – Evaluate (every task in this assignment is at K5 level)

Mandatory Execution and GitHub Submission

- Every student must individually execute every task in this assignment, including all "Complete the Code" programs.
- Create a GitHub repository named DSA01-<RegisterNumber>-Assignment2 and share it with the course faculty (public, or private with the faculty added as a collaborator).
- Keep one folder for each task (for example: Task01, Task02 … or Q1, Q2 …), containing the C++ source files, input files and output screenshots.
- Include a README.md with your name, register number, compiler and version (for example g++ 13), the exact g++ command used to compile (for example g++ -std=c++17 -Wall -Wextra), the steps to run each program, and a short note of the result of each task.
- Commit your work progressively – at least one commit for each task. The commit history will be checked; a single upload of all files on the last day will not be accepted.
- submit the GitHub link below and on the cover page of your report. An assignment submitted without a working GitHub link will not be evaluated.

GitHub Repository Link: _____

MARK DISTRIBUTION

Question	Assessed Area	Bloom's Level	Marks
Q1	Call by value, reference and pointer	K5 – Evaluate	6
Q2	Return by reference	K5 – Evaluate	4

Q3	Default arguments and function overloading	K5 – Evaluate	6
Q4	Class specification with arrays within a class	K5 – Evaluate	6
Q5	Static members and arrays of objects	K5 – Evaluate	7
Q6	Friend and inline functions	K5 – Evaluate	8
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	K5 – Evaluate	13
	TOTAL		50

Submission Requirements

GitHub repository link (mandatory) containing the executed code for every task, including every completed "Complete the Code" program, with input files, output screenshots and a README.md.

- Your assignment report containing the required evidence for Q1–Q7
- C++ source code for all programs
- Predictions, outputs, timings and compiler messages used as evidence

Assessment Verification

Students may be asked to explain any part of their predictions, code, tests or results during a brief verification discussion/viva.

Overall Assessment Expectation

Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate